LEARNMATE – AI STUDYBUDDY

6. Project Testing Phase

Testing authentication, material handling, AI services, database operations and REST APIs.

Date	21-09-26
Team ID	SWTID-2026-5714
Project Name	LearnMate - AI StudyBuddy
Maximum marks	4 Marks



Allows users to upload, retrieve, update, and manage study materials.

3. AI Feature APIs

Provides endpoints for:

- AI Summary Generation
- AI Flashcard Generation
- AI Quiz Generation
- AI Study Plan Generation

4. Administrative APIs

Provides administrator-only endpoints for managing users and monitoring system activities.

Epic 5: Project Execution

The AI StudyBuddy backend can be executed locally using Node.js. Before running the application, ensure that MongoDB is running and the required environment variables are configured correctly. The backend exposes REST APIs that can be tested using Postman or Thunder Client.

Running the Backend Server

Install all required dependencies:

```
npm install
```

Start the development server:

```
npm start
```

After successful execution, the server connects to MongoDB and starts listening for incoming API requests.

API Testing

The backend APIs can be tested using **Postman**.

1. User Registration

Registers a new user by accepting the required details and securely storing the encrypted password in the MongoDB database.

Endpoint:

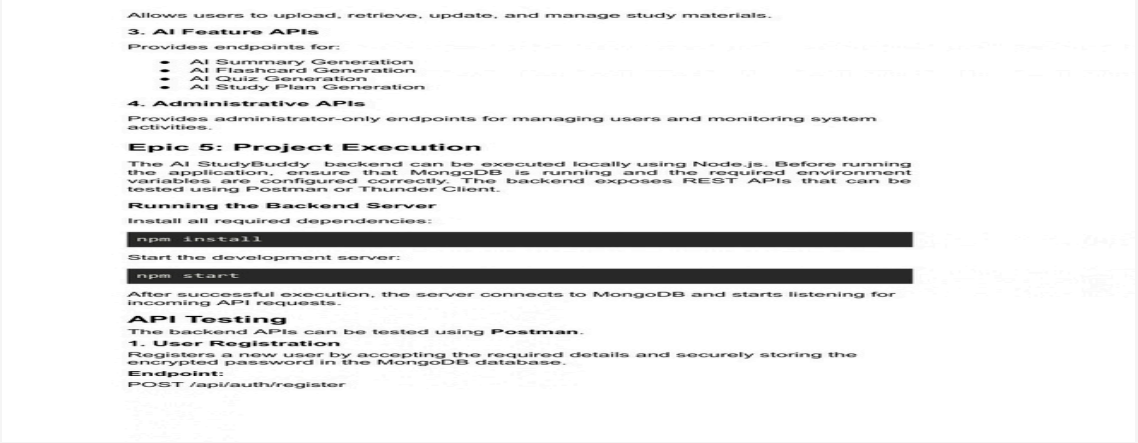
POST /api/auth/register

AI-powered learning assistance and smart study support platform

Testing Approach

Key Points

- Run the backend locally after installing dependencies and configuring MongoDB and environment variables.
- Use Postman or Thunder Client to verify REST API behavior and responses.
- Check protected access through JWT authentication and role-based authorization.



Allows users to upload, retrieve, update, and manage study materials.

3. AI Feature APIs

Provides endpoints for:

- AI Summary Generation
- AI Flashcard Generation
- AI Quiz Generation
- AI Study Plan Generation

4. Administrative APIs

Provides administrator-only endpoints for managing users and monitoring system activities.

Epic 5: Project Execution

The AI StudyBuddy backend can be executed locally using Node.js. Before running the application, ensure that MongoDB is running and the required environment variables are configured correctly. The backend exposes REST APIs that can be tested using Postman or Thunder Client.

Running the Backend Server

Install all required dependencies:

```
npm install
```

Start the development server:

```
npm start
```

After successful execution, the server connects to MongoDB and starts listening for incoming API requests.

API Testing

The backend APIs can be tested using **Postman**.

1. User Registration

Registers a new user by accepting the required details and securely storing the encrypted password in the MongoDB database.

Endpoint:

POST /api/auth/register

Reference visual from the supplied LearnMate project document.

- Verify uploaded material is accepted and passed to AI-powered generation services.

API Test Cases

Key Points

- User Registration: POST /api/auth/register.
- User Login: POST /api/auth/login.
- Upload Study Material: POST /api/material/upload.
- Generate AI Summary: POST /api/materials/:id/summarize.
- Generate AI Flashcards: POST /api/ai/flashcards.
- Generate AI Quiz: POST /api/ai/quiz.
- Generate AI Study Plan: POST /api/ai/study-plan.

Testing Goal

Key Points

- Confirm that authentication, database connectivity, study-material handling and AI-generated resources work through the defined REST interfaces.